Introduction to Neutrino Cosmology

DeMille Group Junior Graduate Students Journal Club

Geoffrey Zheng
University of Chicago

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Foreword

This is a guest lecture on neutrino cosmology given by me at the DeMille group journal club for junior graduate students in the group. Since a primary theme of the DeMille group's research is probing physics beyond the Standard Model (BSM), it is important for all group members to be familiar with the disparate approaches to this rich field of physics research. This particularly includes studying fundamental/BSM physics through the lens of cosmology and astroparticle physics. Neutrino physics is a fascinating subject because although neutrinos are a well-established part of the Standard Model, they may also be sensitive probes of BSM physics. Hence it is worthwhile to understand the basics of neutrino physics from the point of view of cosmology and astroparticle physics.

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1 Introduction

A primary theme of the DeMille group's research is probing physics beyond the Standard Model (BSM). Therefore, it is important for group members to be familiar with the disparate approaches to this rich field of physics research. This includes understanding fundamental/BSM physics through the lens of cosmology and astroparticle physics. Neutrino physics is a fascinating subject because although neutrinos are a well-established part of the Standard Model, they are also sensitive probes of BSM physics. Hence it is worthwhile to understand the basics of neutrino physics from the point of view of cosmology and astroparticle physics.

2 Neutrinos in Particle Physics

2.1 Solving a Problem in Beta Decay

The neutrino (from Italian for "little neutral one") was first proposed as a hypothetical particle by the physicist Wolfgang Pauli in 1930 as a mechanism to explain how energy and angular momentum could remain conserved in beta decay. For instance, consider the following β^- decay that is relevant for dating ancient biological samples (carbon-14 dating):

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + \mathrm{e}^- + \bar{\nu}_e. \tag{2.1}$$

Before physicists knew that neutrinos existed, they expected the resultant beta particle (electron) to have a narrow energy distribution since it should carry away the energy difference between initial and final nuclear states. However, energy measurements actually revealed a continuous spectrum. This seemed to imply a violation of conservation of energy. Another problem was that Carbon-14 nucleus has integer spin quantum number, which is true in general for nuclei with even atomic mass number. Since Nitrogen-14 also has even atomic mass number, its nuclear spin is also an integer. But the electron has spin 1/2. So this also seemed to imply a violation of conservation of angular momentum. These problems motivated Pauli to introduce the neutrino as a "hidden particle" that was affecting the energy spectrum of the emitted electron, and which also had spin 1/2.

Although Pauli strongly believed that the particle existed, he doubted that it could ever be detected experimentally because it had to have an extremely small mass and be electrically neutral [1]. However, less than 30 years later, Clyde Cowan and Frederick Reines announced the direct detection of electron antineutrinos through their interaction with the protons of water molecules in a large water tank near a nuclear reactor [2]. Reines received the 1995 Nobel Prize in Physics for his and Cowan's experimental discovery of the neutrino. Moreover, by the 1970s, neutrinos were fully integrated into the Standard Model of particle physics.

2.2 Neutrinos in the Standard Model

In the SM, neutrinos are electrically neutral (and massless!) leptons that only interact with other particles via the weak force and gravity [3]. There are three flavors of neutrinos, which correspond to the three generations of charged leptons: the electron neutrino ν_e, the muon neutrino ν_μ, and the tau neutrino ν_τ; each neutrino flavor also has an associated antimatter particle called an antineutrino. Furthermore, because parity is violated in weak interactions, helicity is a meaningful quantum number for neutrinos. Qualitatively, helicity is defined as the sign of the projection of a particle's spin angular momentum vector onto its linear momentum vector [3]. All three known

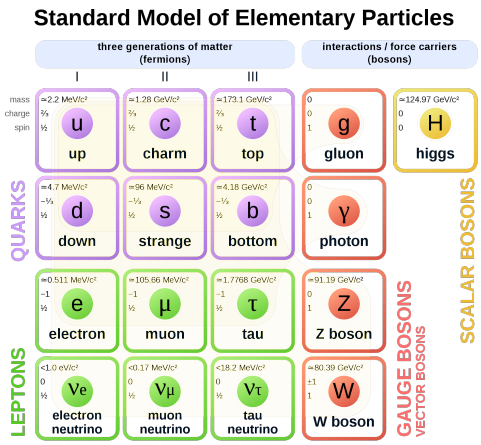

Figure 1: Cartoon schematic of the known elementary particles in the Standard Model of particle physics. Neutrinos come in three flavors (electron, muon, tau) and constitute three of the six leptons. Antimatter particles not shown. Figure from Wikimedia Commons.

flavors of neutrinos are left-handed (helicity $= -1$), while all three known flavors of antineutrinos are right-handed (helicity $= +1$).

The SM, and in particular the electroweak theory, provides a mechanism for explaining the beta decay of carbon-14. In that specific case, it is due to the beta decay of one of the neutrons in the nucleus:

$$\mathrm{n} \to \mathrm{p}^+ + \mathrm{e}^- + \bar{\nu}_e. \tag{2.2}$$

The associated Feynman diagram is shown in Figure 2 below:

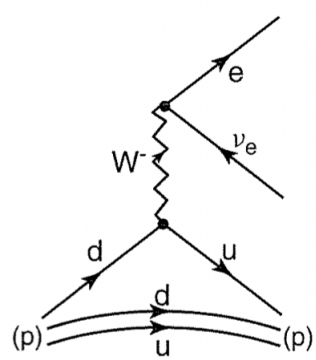

Figure 2: Beta decay of the neutron due to the weak charged current.

This Feynman diagram points out a unique property of the weak interaction compared to the electromagnetic and strong interactions. Namely, only the weak interaction (via the weak charged-current, i.e. exchange of $W^{\pm}$ bosons) allows for change of flavor between particles at a particular vertex. We see this above at both vertices, where a down quark from the neutron changes into an up quark, and an electron antineutrino changes into an electron.

Canonically, charged weak interactions between quarks versus between leptons differ in nature.

In the case of the quarks, the weak interaction does not fully respect the quark flavor states, so that it is possible for cross-generational quark flavor changing to occur (i.e. weak interaction deals with d', s', b' quark states which are in a rotated basis from d, s, b). This is codified in the Cabibbo-Kobayashi-Maskawa (CKM) matrix, which is also related to CP violation. In the case of the leptons, the weak interaction *nominally* respects the lepton flavor states (i.e. lepton number is conserved). This means that flavor changing can only be intra-generational and not inter-generational (i.e. weak interaction can only change electron to electron neutrino, not a muon neutrino). However, as I will discuss below, the experimental discovery of neutrino oscillations made it possible for leptons to undergo inter-generational flavor change. Hence, there also exists a PMNS matrix (analogous to CKM but for leptons) which codifies how lepton flavor can change across generations. However, since neutrino oscillations cannot be explained by SM formalism, this is not a canonical part of the theory (yet).

2.3 Neutrino Oscillation Experiments

Neutrino oscillations were first considered to be possible by Pontecorvo in the late 1950s. Further hints of its possibility arose through the solar neutrino problem in the 1960s, whereby the Homestake experiment detected a missing flux in neutrinos from the Sun (was only sensitive to electron neutrinos, and so always found a result that was 1/3 of the theoretical prediction by Bahcall). Finally, in the late 1990s, terrestrial experiments at Super-Kamiokande Observatory (Japan) and Sudbury Neutrino Observatories (Canada) detected the phenomenon of neutrino oscillations for the first time, which directly implied that neutrinos have non-zero mass [4, 5]. Through solar and atmospheric neutrino experiments, the mass splitting values between the three neutrino species has now been determined relatively precisely. The most recent values quoted in the 2018 Particle Physics Review are [6]:

$$\Delta m_{12}^2 \equiv m_2^2 - m_1^2 = 7.37 \times 10^{-5}\text{eV}^2. \tag{2.3}$$

$$|\Delta m_{13}^2| \equiv |m_3^2 - (m_1^2 + m_2^2)/2| = 2.525 \times 10^{-3}\text{eV}^2. \tag{2.4}$$

Terrestrial neutrino experiments can currently only measure the square differences of the three masses; the absolute sum of the neutrino masses cannot be exactly determined from these measurements alone. Furthermore, because we only know the absolute value of $|\Delta m_{13}^2|$, two possible neutrino mass hierarchies arise. The first hierarchy is known as the normal hierarchy, and assumes that $m_1 \ll m_2 < m_3$. The second hierarchy is known as the inverted hierarchy, and assumes $m_3 \ll m_1 < m_2$. The most recent results from the Super-Kamiokande experiment weakly favor the normal hierarchy [7]. It is anticipated that the results from the ongoing DUNE and Project 8 experiments will provide a more definitive result of the mass ordering of neutrino species [8,9]. According to the normal hierarchy, the lower bound on the total sum of neutrino mass $m_\nu \equiv m_1 + m_2 + m_3$ is predicted to be $m_\nu \gtrsim 0.06$ eV, while according to the inverted hierarchy, the lower bound is predicted to be $m_\nu \gtrsim 0.1$ eV [6]. Upper limits can be obtained terrestrially via beta decay of tritium, or astrophysically from Supernova 1987A (I will not go into those details here).

Here, it will be instructive to go through the basic exercise of deriving how neutrino oscillations work. For simplicity, assume there are only two flavors and two nonzero masses of neutrinos (e.g. electron and muon, they're the longest-lived leptons anyway). It is known that the neutrinos produced in weak interactions are flavor eigenstates, which we will denote $|\nu_e\rangle$ and $|\nu_\mu\rangle$ (for electron and muon neutrino). Flavor eigenstates do not have well-defined mass, but are instead the superposition of two mass eigenstates $|\nu_1\rangle$ and $|\nu_2\rangle$ that have definite masses m_1 and m_2. In the 2-D

Hilbert space of mass eigenstates, the flavor eigenstates make an angle θ (called the mixing angle) with the mass eigenstates:

$$\begin{pmatrix} |\nu_e\rangle \\ |\nu_\mu\rangle \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} |\nu_1\rangle \\ |\nu_2\rangle \end{pmatrix}. \tag{2.5}$$

The two mass eigenstates have well-defined energies $E_{1,2}$ given by:

$$E_{1,2}(p) = \left(p^2c^2 + m_{1,2}^2c^4\right)^{1/2}, \tag{2.6}$$

where we use the relativistic formula for energy because neutrinos travel extremely fast. The two flavor eigenstates do not have well-defined energies. Neutrinos travel freely through space as mass/energy eigenstates.

First, we calculate the time evolution of a state $|\psi(t)\rangle$ that is a pure electron neutrino at $t = 0$, i.e. $|\psi(0)\rangle = |\nu_e\rangle$. We wish to express our answer in terms of mass eigenstates, energies, and the mixing angle. Since the mass eigenstates have definite energy, the Hamiltonian is diagonal in that basis. So we first need to express $|\nu_e\rangle$ in terms of the mass eigenstates. From the mixing angle matrix, we find:

$$|\psi(0)\rangle = |\nu_e\rangle = \cos\theta\,|\nu_1\rangle - \sin\theta\,|\nu_2\rangle\,. \tag{2.7}$$

Hence, the state evolves under unitary time evolution by the Hamiltonian to become:

$$|\psi(t)\rangle = \cos\theta\, e^{-iE_1t/\hbar}\,|\nu_1\rangle - \sin\theta\, e^{-iE_2t/\hbar}\,|\nu_2\rangle\,. \tag{2.8}$$

Next, we calculate the probability $P(\nu_e \to \nu_\mu)$ of a transition from $|\nu_e\rangle$ to $|\nu_\mu\rangle$, and show that it is periodic in time. We wish to express our answer in terms of $E_{1,2}$ and θ. First, we need to convert $|\psi(t)\rangle$ back to the flavor eigenstate basis. Using the mixing angle matrix again, we find:

$$|\nu_1\rangle = \cos\theta\,|\nu_e\rangle + \sin\theta\,|\nu_\mu\rangle\,, \quad |\nu_2\rangle = -\sin\theta\,|\nu_e\rangle + \cos\theta\,|\nu_\mu\rangle\,. \tag{2.9}$$

Hence:

$$|\psi(t)\rangle = \cos\theta\, e^{-\frac{iE_1t}{\hbar}}\left(\cos\theta\,|\nu_e\rangle + \sin\theta\,|\nu_\mu\rangle\right) - \sin\theta\, e^{-\frac{iE_2t}{\hbar}}\left(-\sin\theta\,|\nu_e\rangle + \cos\theta\,|\nu_\mu\rangle\right) \tag{2.10}$$

$$= \left(\cos^2\theta\, e^{-\frac{iE_1t}{\hbar}} + \sin^2\theta\, e^{-\frac{iE_2t}{\hbar}}\right)|\nu_e\rangle + \cos\theta\sin\theta\left(e^{-\frac{iE_1t}{\hbar}} - e^{-\frac{iE_2t}{\hbar}}\right)|\nu_\mu\rangle\,. \tag{2.11}$$

So the probability of the neutrino being in the $|\nu_\mu\rangle$ eigenstate at time t is:

$$P(\nu_e \to \nu_\mu) = \left|\cos\theta\sin\theta\left(e^{-\frac{iE_1t}{\hbar}} - e^{-\frac{iE_2t}{\hbar}}\right)\right|^2 \tag{2.12}$$

$$= \frac{1}{2}\sin^2(2\theta)\left(1 - \cos\left(\frac{E_2 - E_1}{\hbar}t\right)\right) \tag{2.13}$$

$$\implies P(\nu_e \to \nu_\mu) = \sin^2(2\theta)\sin^2\left(\frac{E_2 - E_1}{2\hbar}t\right). \tag{2.14}$$

Hence the probability oscillates in time, with amplitude modulated by the mixing angle.

Finally, we wish to obtain an expression for the spatial period of neutrino oscillation λ as a function of average neutrino energy $E_\nu = pc$, mass square difference $\Delta m^2 \equiv m_2^2 - m_1^2$, and flight distance $L = ct$. We can assume the rest masses of the two mass eigenstates are much smaller than

their momenta, i.e. $m_{1,2} \ll p/c$, and just use the lowest order contribution to the energy due to non-zero neutrino mass.

Since the masses are much smaller than the momenta, to first order we have:

$$E_{1,2} \approx pc\left(1 + \frac{m_{1,2}^2 c^2}{2p^2}\right). \tag{2.15}$$

If we call the average energy of the neutrino $E_\nu = pc$, then the difference in energy is:

$$E_2 - E_1 = \frac{(m_2^2 - m_1^2)pc^3}{2p^2} = \frac{\Delta m^2 c^4}{2E_\nu}. \tag{2.16}$$

Recalling that the flight distance of the neutrino is simply $L = ct$, the expression for $P(\nu_e \to \nu_\mu)$ becomes:

$$P(\nu_e \to \nu_\mu) = \sin^2(2\theta)\,\sin^2\left(\Delta m^2 c^4 \frac{L}{4E_\nu \hbar c}\right). \tag{2.17}$$

The period of a $\sin^2(x)$ function is π. Hence, the spatial period of oscillations is:

$$\lambda = \frac{4\pi E_\nu \hbar c}{\Delta m^2 c^4}. \tag{2.18}$$

Hence experimentalists can thus measure λ to deduce the square difference in neutrino masses Δm^2. Notably, neutrino oscillations directly imply that the masses of the neutrinos are different (so in other words, they can't all be 0). Extrapolating to the real-life case of 3 neutrino species, it follows that by measuring Δm_{12}^2 and $|\Delta m_{13}^2|$ (which is agnostic to NO vs IO), we can obtain lower bounds on $\sum m_\nu$ via a "Lagrange multiplier"-type argument (i.e. given that $x^2 - y^2 = c$, $x^2 - z^2 = d$, constrain the minimum value of $x + y + z$.

3 Neutrinos in Cosmology

Much of the role that neutrinos play in cosmology can be understand in terms of the thermodynamics of Standard Model particles during the early Universe. We first go through a general analysis of early Universe thermodynamics before applying it to the specific case of neutrinos.

3.1 Early Universe Thermodynamics

The cosmic history of our Universe can be described as a transition from the incredibly hot and dense state right after the Big Bang to the incredibly cool and dilute space that we find ourselves in today. In the early Universe, particles interacted so frequently with one another that they maintained a constant temperature, thereby existing in a state of thermodynamic equilibrium known as the "thermal bath". As the Universe expanded and cooled, particles interacted less frequently with each other and gradually fell out of equilibrium, thereby decoupling from the thermal bath.

Three quantities of particular interest to us for any particle species are: number density n, energy density ρ, and pressure p. We will always consider them in one of two limits:

- Relativistic limit: $T \gg m$. Recall from Einstein's theory of relativity that rest energy $\sim m$, while kinetic energy $\sim T$. Hence when kinetic energy is much greater than rest energy, the particle is moving extremely fast and is relativistic.

- Non-relativistic limit: $T \ll m$.

In the early Universe, T was much higher than the rest mass of any Standard Model particle. So all particles in the thermal bath in the early Universe could be considered relativistic. Moreover, we typically can neglect the effect of chemical potential μ. In this case, using the Fermi-Dirac and Bose-Einstein distributions from statistical mechanics in the limit of $T \gg m, \mu$, we have:

$$n = \begin{cases} \frac{\zeta(3) g T^3}{\pi^2}, & \text{boson} \\ \frac{3}{4}\frac{\zeta(3) g T^3}{\pi^2}, & \text{fermion} \end{cases} \tag{3.1}$$

$$\rho = \begin{cases} \frac{\pi^2 g T^4}{30}, & \text{boson} \\ \frac{7}{8}\frac{\pi^2 g T^4}{30}, & \text{fermion} \end{cases} \tag{3.2}$$

$$p = \begin{cases} \frac{\pi^2 g T^4}{90}, & \text{boson} \\ \frac{7}{8}\frac{\pi^2 g T^4}{90}, & \text{fermion.} \end{cases} \tag{3.3}$$

Here, g counts the number of internal degrees of freedom of the particle species (i.e. its "degeneracies"). This includes all possible spin states as well as any states that exist with different charges or other quantum numbers which would not affect the number density, energy density, or pressure. On the other hand, in the non-relativistic limit $T \ll m$, the above expressions reduce to the classical Maxwell-Boltzmann distribution:

$$n = g\left(\frac{mT}{2\pi}\right)^{3/2} e^{-(m-\mu)/T} \tag{3.4}$$

$$\rho = g m^{5/2}\left(\frac{T}{2\pi}\right)^{3/2} e^{-(m-\mu)/T} \tag{3.5}$$

$$p = g T^{5/2}\left(\frac{m}{2\pi}\right)^{3/2} e^{-(m-\mu)/T}. \tag{3.6}$$

In practice in cosmology, we care about the energy density of an ensemble of species (basically, all the species in the Standard Model). In this case, we can just sum up the energy densities of each individual species and write the total energy density of the ensemble as:

$$\rho = \sum_i \rho_i \equiv \frac{\pi^2 g_\star(T) T^4}{30}, \tag{3.7}$$

where the sum is over the particle species and T is the temperature of the photons in the thermal bath (since photons are massless, so they will always remain relativistic). Here $g_\star(T)$ is defined as the number of relativistic degrees of freedom in the ensemble; it is a function of temperature because as the Universe cools and expands, more and more particles fall out of equilibrium with the thermal bath.

In the case where all particle species in question are highly relativistic ($T_i \gg m_i$ for all i), we can also write the total energy density of the ensemble as:

$$\rho = \sum_{i=\text{bosons}} \frac{\pi^2 g_i T_i^4}{30} + \sum_{i=\text{fermions}} \frac{7}{8}\frac{\pi^2 g_i T_i^4}{30} \tag{3.8}$$

$$= \frac{\pi^2 T^4}{30}\left[\sum_{i=\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8}\sum_{i=\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4\right], \tag{3.9}$$

where T is the photon temperature, T_i is the temperature of species i, and g_i is the number of internal degrees of freedom of species i. Comparing equation 3.9 with equation 3.7, we see that if all particle species are in kinetic equilibrium ($T_i = T$) and are highly relativistic ($m_i \ll T$), then:

$$g_\star = \sum_{i=\text{bosons}} g_i + \frac{7}{8}\sum_{i=\text{fermions}} g_i. \tag{3.10}$$

Besides n, ρ, and p, the other useful thermodynamic quantity to consider while analyzing the thermal history of the Universe is the entropy density s. It is defined as:

$$s \equiv \frac{S}{V} = \frac{\rho + p}{T}, \tag{3.11}$$

where the second equality follows from the 1st Law of Thermodynamics ($dE = T\,dS - p\,dV \implies E = TS - PV$ and hence $S/V = E/VT + P/T = (\rho + p)/T$. Just as above (equation 3.7), we can write the total entropy density in terms of the photon temperature T and a quantity $g_{\star,S}$ defined as the number of relativistic degrees of freedom in entropy:

$$s \equiv \frac{2\pi^2 g_{\star,S}(T) T^3}{45}. \tag{3.12}$$

Note the $2\pi^2/45$ prefactor can be seen from simply adding the prefactors from ρ and p, and the temperature dependence scales as T^3 instead of T^4. s is typically dominated by the relativistic particles in the thermal bath, so we recall from the relativistic equation of state that $p = \rho/3$. Hence $s = 4\rho/3T$. So again, in the highly relativistic limit $T_i \gg m_i$ for all i, we have that:

$$s = \frac{4}{3}\frac{\rho}{T} = \frac{4}{3}\left[\sum_{i=\text{bosons}} \frac{\pi^2 g_i T_i^3}{30} + \frac{7}{8}\sum_{i=\text{fermions}} \frac{\pi^2 g_i T_i^3}{30}\right]. \tag{3.13}$$

Comparing equation 3.12 with equation 3.13, we obtain an expression for $g_{\star,S}$ for highly relativistic particles:

$$g_{\star,S} = \sum_{i=\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8}\sum_{i=\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3. \tag{3.14}$$

Note that $g_{\star,S}$ is similar to $g_\star$ except that the former scales as $(T_i/T)^3$ instead of $(T_i/T)^4$.

Entropy is a useful thermodynamic variable in the thermal history of the early Universe because to very good approximation, the expansion of the early Universe was adiabatic. This means:

$$dQ = TdS = 0, \tag{3.15}$$

i.e. entropy was a conserved quantity. In an expanding universe, this means the amount of entropy in a given comoving volume remains constant as space expands. Hence:

$$S = sV \propto sa^3 = \text{constant} \quad \implies s \propto a^{-3}. \tag{3.16}$$

Combining this with equation 3.12, this means that $g_{\star,S}T^3a^3$ remains constant as space expands. Equivalently, the temperature of the thermal bath (i.e. the relic photons at present day) should evolve like:

$$T \propto \frac{1}{ag_{\star,S}^{1/3}}. \tag{3.17}$$

Note that $g_{\star,S}$ is a step function; that is, it remains constant for a certain temperature range, until $T \sim m_i$ for any particle species of interest i. So generically, we see that $T \sim 1/a$. Once the temperature approaches the mass of one or more particle species in the bath, those species begin to be destroyed more often than they are created, reducing their abundance and transferring their energy into lighter particle species. This transfer of energy is known as an "entropy dump."

Up until now, we have discussed the equilibrium thermodynamics of the early Universe. But we know that the various particle species of the Standard Model must have fallen out of equilibrium at some point (otherwise we would not exist in our current form today). Intuitively, we know that equilibrium can be maintained whenever the relevant interactions between particles occur at a much faster rate than the rate at which the background environment is changing. In terms of cosmology, this means that departure from equilibrium occurs when the rate of scattering between particles of interest in the bath falls below the rate of Hubble expansion (i.e. the Universe is literally growing so fast the particles can't find other particles to interact with anymore). Quantitatively, the equilibrium condition can be expressed as:

$$\Gamma = n\sigma v \gtrsim H, \tag{3.18}$$

where n is number density, σ is scattering cross section, v is relative velocity, and H is Hubble parameter. Below, we will apply the general thermodynamics principles derived above to the specific case of neutrinos.

3.2 Decoupling of Neutrinos from the Thermal Bath

Neutrinos only interact with other Standard Model particles via the weak force. Therefore, neutrinos stay in equilibrium with the thermal bath of the early Universe through exchange of W and Z bosons. For simplicity, let us assume $T \ll m_\mu$, i.e. the temperature of the thermal bath has dropped well below the mass of the muon, so that we only care about interactions in the 1st generation of leptons. Namely, the following processes would still be allowed: s-channel Z exchange or t-channel W exchange of ν_e and $\bar{\nu}_e$ annihilation, and s-channel W exchange of ν_e and e^- scattering.

We calculate the neutrino decoupling temperature by looking at the ratio of the neutrino-thermal bath interaction rate to the Hubble expansion. To very good approximation, the neutrinos are still so relativistic at this time that $v \approx c$; hence the scattering rate of the neutrinos is $\Gamma = n_\nu \sigma c$ (even though we know neutrinos are massive, they don't become non-relativistic until very late because of how light they are). On the other hand, from the 1st Friedmann equation, $H = (8\pi G\rho/3)^{0.5}$. So we find:

$$\frac{\Gamma}{H} = \frac{n_{e,\nu}\sigma c}{(8\pi G\rho/3)^{0.5}} \approx \left(\frac{10.75}{g_\star(T)}\right)^{1/2} \left(\frac{T}{3\,\text{MeV}}\right)^3. \tag{3.19}$$

Hence when $T \lesssim 3\,\text{MeV}$, H starts to exceed Γ and the neutrinos decouple from the thermal bath. Upon decoupling, the temperature of the neutrinos still remained approximately the same as that of the electrons/positrons and photons (only remaining particles in the bath), since both scale as $\sim 1/a$ due to their relativistic nature and there is not an immediate change in $g_\star$. But this did not remain true for long, because electron-positron annihilation occurred roughly also around $T \sim 1$ MeV (t ~ 1 second after Big Bang). Electron-positron annihilation is given by the formula:

$$e^+ + e^- \leftrightarrow \gamma + \gamma. \tag{3.20}$$

Once the temperature of the bath drops below $T \sim 1$ MeV, the reverse reaction does not occur as rapidly, and hence nearly all electrons and positrons annihilated into photons. This only left relic photons in the thermal bath. To summarize:

- After neutrinos have decoupled, they are at temperature T_ν. For simplicity, we can assume that electron-positron annihilation took place very soon after neutrino decoupling, so T_ν is still approximately the temperature of the thermal bath. The only remaining relativistic species in the bath prior to annihilation are e^-, e^+, and γ (all have 2 spin degrees of freedom), so:

$$g_{\star,S} = 2 + \frac{7}{8}(2 \times 2) = \frac{11}{2}.$$

- After electrons and positrons have decoupled from annihilation, the only remaining relativistic species is photons, so:

$$g_{\star,S} = 2.$$

Now, using the conservation of entropy in the thermal bath (both prior to electron-positron annihilation and afterward), we have:

$$g_{\star S,i}(a_i T_\nu)^3 = g_{\star S,f}(a_f T_\gamma)^3 \tag{3.21}$$

$$\implies \frac{11}{2}(a_i T_\nu)^3 = 2(a_f T_\gamma)^3. \tag{3.22}$$

Since we assume electron-positron annihilation was approximately instantaneous, we can take $a_i \approx a_f$. Solving for T_ν in terms of T_γ thus gives:

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma. \tag{3.23}$$

T_ν represents the temperature of the relic neutrinos (i.e. the CνB) ever since they decoupled from the thermal bath. T_γ represents the temperature of the relic photons, which are the CMB.

3.3 A Crude Upper Bound on the Neutrino Mass from Cosmology

From special relativity, we know that mass and energy density are related. Moreover, from non-relativistic statistical mechanics, we can obtain relations between number density and energy density. Hence, we can use our formalism to obtain a crude upper bound on the sum of neutrino masses from cosmology. Note that this will be somewhat hand-waving because we will treat neutrinos as non-relativistic (i.e. temperature = kinetic energy < rest mass), but since relic neutrinos still travel around $\lesssim 0.1c$, we will sometimes still treat them quasi-relativistically.

We start by computing the number density of relic neutrinos today. Although neutrinos are non-relativistic today, they still travel fast enough that we will still use the relativistic Fermi-Dirac distribution:

$$n_{\nu,0} = \frac{3}{4}\frac{\zeta(3)}{\pi^2} g T_{\nu,0}^3. \tag{3.24}$$

We calculate $T_{\nu,0}$, the temperature of neutrinos today, using the fact that we know CMB photons today have temperature $T_{\gamma,0} = 2.73$ K. So:

$$T_{\nu,0} = \left(\frac{4}{11}\right)^{1/3} 2.73\,\mathrm{K} = 1.949\,\mathrm{K} = 0.168\,\mathrm{meV}. \tag{3.25}$$

Neutrinos are spin-1/2 fermions, so $g = 2$. Therefore:

$$n_{\nu,0} = \frac{3}{2}\frac{\zeta(3)}{\pi^2}(0.168\,\mathrm{meV})^3 = 1.126 \times 10^8\,\mathrm{neutrinos/m^3} \approx 113\,\mathrm{neutrinos/cm^3}. \tag{3.26}$$

Finally, to calculate the energy density contribution of the massive neutrino species, we recall that they will be non-relativistic today, so their energy density is given by (as can be seen by equations 3.4 and 3.5):

$$\rho_{\nu,0} = m_\nu n_{\nu,0}. \tag{3.27}$$

Dividing out by the critical density of the universe today[1] gives:

$$\frac{\rho_{\nu,0}}{\rho_{\mathrm{crit},0}} = \Omega_\nu = \frac{m_\nu n_{\nu,0}}{\rho_{\mathrm{crit},0}}. \tag{3.28}$$

We compute the critical density in terms of h as follows[2]:

$$\rho_{\mathrm{crit},0} = \frac{3H_0^2}{8\pi G} = \frac{3 \times 10^4\, h^2\,(\mathrm{km/s/Mpc})^2}{8\pi \times 6.67 \times 10^{-11}\,\mathrm{m^3/kg/s^2}} = 1.88 \times 10^{-26} h^2\,\mathrm{kg/m^3} = 1.06 \times 10^{10} h^2\,\mathrm{eV/m^3}. \tag{3.29}$$

Hence, the neutrino contribution to the energy density is:

$$\Omega_\nu = \frac{(1.126 \times 10^8\,\mathrm{neutrinos/m^3})\, m_\nu}{1.06 \times 10^{10} h^2\,\mathrm{eV/m^3}} \quad \Longrightarrow \quad \Omega_\nu h^2 \approx \frac{m_\nu}{94\,\mathrm{eV}}. \tag{3.30}$$

Since we have non-relativistic massive neutrinos, we treat them as matter, not radiation. We know from Planck data that $\Omega_m h^2 = 0.14$, so assuming that all other Ω parameters stay constant in order to keep the universe flat, the maximum bound on the sum of neutrino masses comes from the value of $\Omega_m h^2$ (this would be equivalent to assuming that all matter is neutrinos). This gives:

$$\Omega_\nu h^2 = \frac{\sum m_\nu}{94\,\mathrm{eV}} < \Omega_m h^2 = 0.14.$$

Hence the bound on the sum of neutrino masses is:

$$\sum m_\nu < 94\,\mathrm{eV} \cdot 0.14 \quad \Longrightarrow \quad \sum m_\nu < 13.16\,\mathrm{eV}. \tag{3.31}$$

[1] The critical density of the universe is defined as the matter energy density needed to make the spacetime geometry of the universe flat, where we assume that there is negligible contribution from radiation and that dark energy does not exist. It turns out the actual *total* energy density of the universe today (matter + radiation + dark energy) is essentially the critical density, because we have measured the spacetime geometry to be flat.

[2] Cosmologists parameterize the Hubble parameter in terms of h and leave theoretical expressions in terms of h so as to be exact. Of course, from measurements we roughly know that $0.67 < h < 0.74$.

Here I point out that this is only a crude upper bound, obtained from the thermodynamics of relic neutrinos and the fact that the fractional energy density of neutrinos is bounded above by the fractional energy density of all matter. A more strict upper bound on neutrino mass ($m_\nu \lesssim 0.2$ eV) comes from CMB anistropy spectrum measurements.

Finally, as a quick digression, consider the following. In Figure 3 below, I show a numerical plot I created that tracks the fractional energy density of all relevant components of the Universe over the course of cosmic history (from $a \sim 10^{-9}$ to present day). The plot is instructive for differentiating between the radiation-dominated, matter-dominated, and cosmological constant-dominated portions of the history of the Universe. The sharp transitions for $m_\nu = 0.01$ eV and $m_\nu = 0.05$ eV show when those species would have become non-relativistic. The plot is generated simply by numerically computing how $\Omega = \rho/\rho_{\text{crit}}$ evolves over time for the components of interest, since ρ generically depends on $H \equiv \dot{a}/a$.

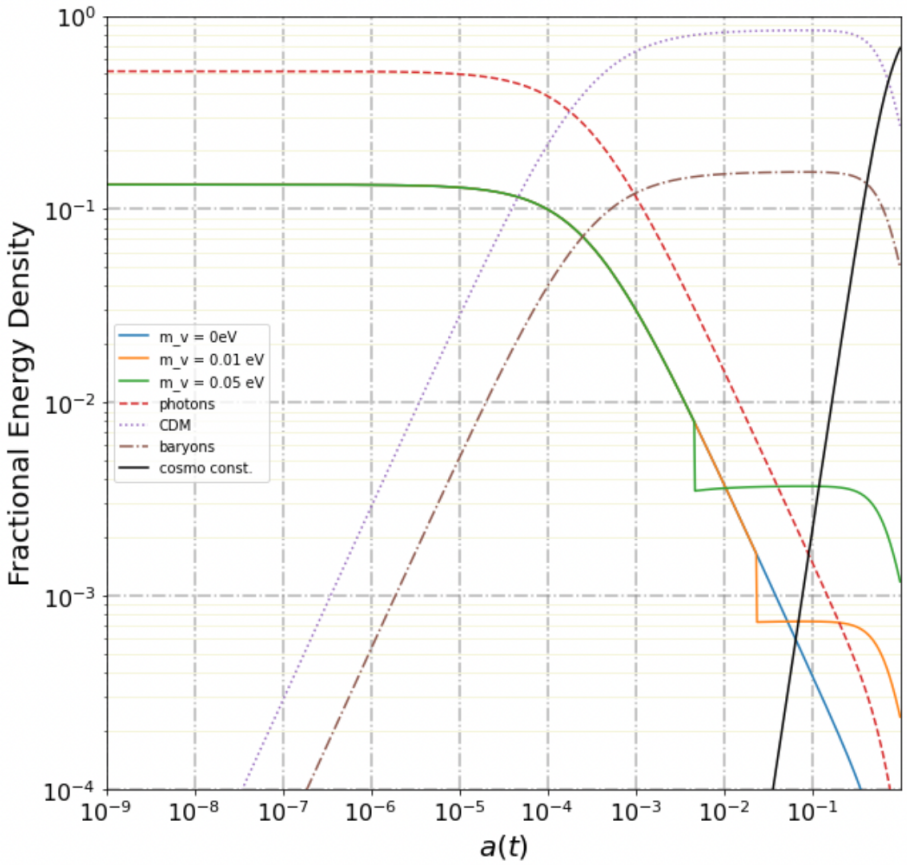

Figure 3: Evolution of the fractional energy density of relevant components of the Universe (photons, neutrinos, CDM, baryons, and cosmological constant).

3.4 Could Neutrinos be Dark Matter?

Before the advent of ΛCDM cosmology as the accepted Standard Model of cosmology, there was a rich debate about the nature of dark matter. In particular, cosmologists used to believe that a significant portion of dark matter (if not all of it) could be "hot", i.e. highly relativistic. Neutrinos

were a viable candidate for being a form of hot dark matter (HDM) because they were extremely light, only interacted via weak force and gravity, and were highly relativistic during the early Universe. Indeed, before experimental verification of neutrino oscillations, most people believed neutrinos were massless, meaning they would always be relativistic.

Here, it is instructive to compute the free-streaming length of an HDM neutrino during the radiation-dominated period (i.e. up to matter-radiation equality). By free-streaming length, we mean the characteristic length scale on which neutrinos can travel before interacting with other particles. But we will be even more hand-wavy and just let the length be the literal distance the neutrino can travel in a certain amount of time, i.e the integral of velocity over time while accounting for scale factor. This is essentially saying the neutrino does not interact with any other particle after decoupling, which is more or less true. We will assume the following: a flat universe with $H_0 = 70$ km/s/Mpc, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $\Omega_r = 9.5 \times 10^{-5}$. Further, we assume that there is only one species of neutrino with mass $m_\nu \sim 30$ eV (roughly the same order of magnitude as the upper limit from cosmology).

We start by computing the scale factor at matter-radiation equality. At matter-radiation equality, their respective energy densities must be equal. So:

$$\rho_m(a_{eq}) = \rho_r(a_{eq}) \quad \Longrightarrow \quad \frac{\Omega_m}{a_{eq}^3} = \frac{\Omega_r}{a_{eq}^4} \quad \Longrightarrow \quad a_{eq} = \frac{\Omega_r}{\Omega_m} = 2.9 \times 10^{-4}. \tag{3.32}$$

Next, we compute the scale factor where our species of massive neutrino becomes non-relativistic. Neutrinos become non-relativistic when their thermal energy is comparable to their rest mass energy. So in other words:

$$3k_B T_\nu \sim m_\nu c^2. \tag{3.33}$$

We recall that:

$$T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma \quad \Longrightarrow T_{\nu,0} = \left(\frac{4}{11}\right)^{1/3} \cdot (2.73\,\mathrm{K}) = 1.95\,\mathrm{K}. \tag{3.34}$$

Relic neutrino temperature scales as $1/a$, so we know that:

$$T_\nu(a) = \frac{T_{\nu,0}}{a}. \tag{3.35}$$

Hence, the scale factor of non-relativistic transition is:

$$a_{nr} = \frac{3k_B T_{\nu,0}}{m_\nu c^2} = 1.68 \times 10^{-5}. \tag{3.36}$$

The corresponding redshifts, using $a = 1/(1+z)$, are: $z_{eq} = 3447$ and $z_{nr} = 59413$. The corresponding times are $t_{nr} \sim 200$ yrs and $t_{eq} \sim 50$ kyrs. So we see that the HDM neutrino becomes non-relativistic far before matter-radiation equality.

Now we can now compute the free-streaming length r_{fs}. It is simply the integral of velocity over time, accounting for the scale factor due to expansion of the Universe. Since the HDM neutrino became non-relativistic far before matter-radiation equality, we split the integral into two parts:

$$r_{\mathrm{fs}} = \int_0^{t_{nr}} \frac{v(t')}{a(t')}\, dt' + \int_{t_{nr}}^{t_{eq}} \frac{v(t')}{a(t')}\, dt'. \tag{3.37}$$

For $0 \leq t \leq t_{nr}$, $v(t') = c$. For $t_{nr} \leq t \leq t_{eq}$, $v(t') = ca_{nr}/a$. We use this formula for velocity in the non-relativistic regime simply to enforce continuity of velocity and length at t_{nr} while maintaining that $v < c$. $a(t') = t'^{1/2}$ throughout because we are in the radiation-dominated period. Plugging in gives:

$$r_{\text{fs}} = \int_0^{t_{nr}} \frac{c}{t'^{1/2}}\, dt' + \int_{t_{nr}}^{t_{eq}} \frac{ca_{nr}}{t'}\, dt' \tag{3.38}$$

$$= 2ct_{nr}^{1/2} + ca_{nr} \ln\left(\frac{t_{eq}}{t_{nr}}\right) \tag{3.39}$$

$$= ct_{nr}\left[2t_{nr}^{-1/2} + a_{nr}t_{nr}^{-1} \ln\left(\frac{t_{eq}}{t_{nr}}\right)\right] \tag{3.40}$$

Recalling that $a_{nr} = t_{nr}^{1/2}$, this gives:

$$r_{\text{fs}} = \left(\frac{ct_{nr}}{a_{nr}}\right)\left[2 + \ln\left(\frac{t_{eq}}{t_{nr}}\right)\right]. \tag{3.41}$$

We calculate t_{nr} and t_{eq} as follows. The universe is radiation-dominated throughout the relevant timescales of this problem, so we know from the Friedmann equation that:

$$\left(\frac{\dot{a}}{a}\right)^2 = H_0^2 \Omega_r a^{-4} \quad \Longrightarrow \quad \frac{da}{dt} = H_0\sqrt{\Omega_r}a^{-1} \tag{3.42}$$

$$\Longrightarrow \int_0^t dt' = \frac{1}{H_0\sqrt{\Omega_r}} \int_0^a a'\, da' \tag{3.43}$$

$$\Longrightarrow t_{eq} = \frac{a_{eq}^2}{2H_0\sqrt{\Omega_r}}, \quad t_{nr} = \frac{a_{nr}^2}{2H_0\sqrt{\Omega_r}}. \tag{3.44}$$

Plugging all these values into the expression for r_{fs} gives:

$$r_{\text{fs}} = \left(\frac{ca_{nr}}{H_0\sqrt{\Omega_r}}\right)\left[1 + \ln\left(\frac{a_{eq}}{a_{nr}}\right)\right] \tag{3.45}$$

$$= \left(\frac{3 \times 10^5\,\text{km/s} \cdot 1.68 \times 10^{-5}}{70\,\text{km/s/Mpc} \cdot \sqrt{9.5 \times 10^{-5}}}\right)\left[1 + \ln\left(\frac{2.9 \times 10^{-4}}{1.68 \times 10^{-5}}\right)\right] \tag{3.46}$$

$$\Longrightarrow r_{\text{fs}} \approx 30\,\text{Mpc}. \tag{3.47}$$

This means that growth of all structures with length scales below 30 Mpc will have been suppressed by neutrinos by the time of matter-radiation equality, so only large galactic superclusters could have remained. Hence, structure formation in a HDM cosmology proceeds in top-down fashion (largest structures form first, followed by smaller ones). Galaxy surveys overwhelmingly indicate that galaxies formed first, followed by galaxy clusters, which largely rules out the top-down structure formation theory. Hence, the theory of neutrinos being HDM was largely discarded in favor of the more canonical cold dark matter (CDM) theory, whose particle nature is still unknown.

It is worth pointing out, however, that this calculation does not rule out neutrinos as *being* HDM. It simply states that even if neutrinos were HDM, it is very unlikely that HDM even makes up a significant proportion of the DM in our Universe. Moreover, it can be seen (but is left as an exercise to the reader to explicitly compute) that if we take a more realistic value for m_ν in the calculation, such as 0.1 eV, then t_{nr} will shift later in time. This is because a lighter neutrino stays

relativistic for longer in time. Since free-streaming length scales roughly linearly in a_{nr}, this would make $r_{\rm fs}$ even larger for lighter m_ν.

Finally, let me discuss some parting thoughts regarding neutrinos, HDM, and the relation between DM and large-scale structure formation in our Universe. First, from cosmological perturbation theory (specifically the Mészaŕos equation and Jeans length), it turns out that DM perturbations scale as follows:

$$\delta_{DM} \sim \ln a, \quad \text{radiation-dominated period} \tag{3.48}$$

$$\delta_{DM} \sim a, \quad \text{matter-dominated period.} \tag{3.49}$$

So DM perturbations cannot really start growing until the matter-dominated regime. On the other hand, baryonic matter cannot even begin to clump until after recombination ($t \sim 380$ kyrs) because of electron-photon scattering prior to that. Cosmological simulations show that if DM didn't exist, then the evolution of baryonic matter clumps after recombination cannot replicate the observed large-scale structure distribution today. Hence, there must exist DM that does not interact with SM particles like photons, so that the DM perturbations could begin clumping from the beginning of the matter-dominated regime. In other words, DM forms a scaffolding for the baryonic matter to later fall into and grow.

From our calculation above, we found that neutrino HDM suppresses clumping below scales of $r_{\rm fs} \sim 30$ Mpc (galactic supercluster size). So if DM were predominantly hot, our Universe would've started from gigantic scaffolding for superclusters, which would've later fragmented into smaller structures like galaxies and stars. Since this is not what happened according to astrophysical observations, we believe most (if not all) DM is cold (non-relativistic from the beginning). CDM should have small free-streaming length during the radiation-domination regime. Two prime candidates for CDM are WIMPs and axions. WIMPs have masses $\sim$ 100s of GeV (electroweak symmetry breaking scale) while axions are much less massive ($m \sim 10^{-3}$ to 10^{-6} eV). Such light CDM particles can exist without being relativistic in the early Universe because they essentially do not interact with the thermal bath of SM particles, and so do not have to be in thermal equilibrium with them at any time.

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